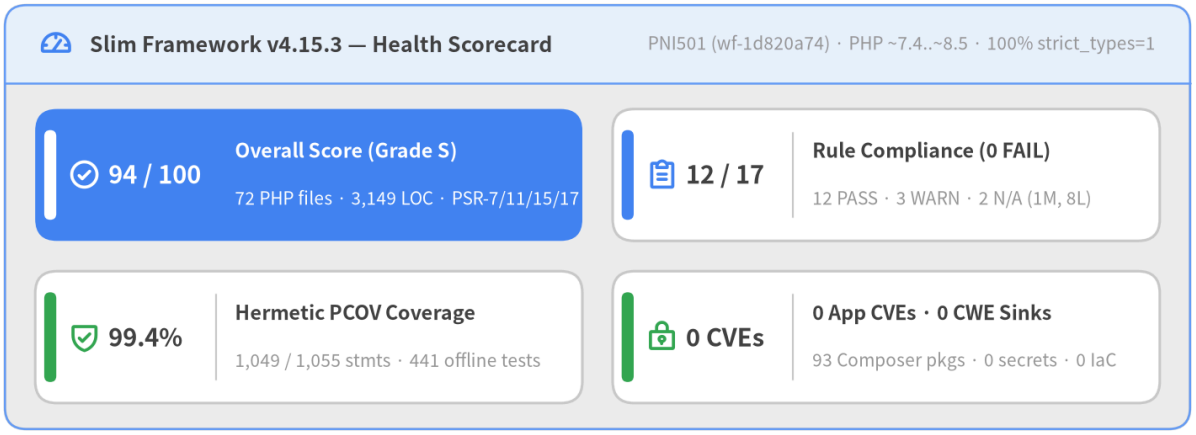
PNI Software Health Diagnostic Report

Slim Framework `v4.15.3` (`slim/slim`) — Automated Static PHP AST, 5-Tier PSR-7/PSR-15 Namespace Dependency DAG, Hermetic PHPUnit + PCOV Container Runtime, and Supply-Chain Security Audit (`pni501` · Run `wf-1d820a74`).



Slim Framework v4.15.3 — PNI501 Overall Engineering & Architecture Health Scorecard (94/100 Grade S)

Diagnostic Attribute	Verified Metadata & Audit Baseline
Target Project	`slim` (`Slim Framework v4.15.3` , package `slim/slim` , PHP `~7.4.0 || ~8.0.0 .. ~8.5.0` , `100%` `declare(strict_types=1)`)
Target Language & Profile	`php` (`library` — Standards-compliant PSR-7, PSR-11, PSR-15 & PSR-17 HTTP micro-framework)
Upstream License & Attribution	MIT License (`LICENSE-SLIM` ; Copyright © Josh Lockhart, Andrew Smith, Rob Allen, Pierre Bérubé, and Slim Framework contributors)
Diagnostic Workflow	`pni501` (Workflow Run ID `wf-1d820a74` , category `audit`)
Evaluated At	`2026-11-10T03:27:24.695841+00:00`
Overall Score & Grade	**94 / 100 (Grade S)** — `12 / 17` rules `PASS` , `3 WARN` , `0 FAIL` , `2 NOT_APPLICABLE` (`9` findings: `0` CRITICAL, `0` HIGH, `1` MEDIUM, `8` LOW)
Analyzed Codebase Scope	`72` PHP source files (`3,149` LOC, `72` classes/interfaces, `311` callables) & `35` PHPUnit test classes (`306` test methods, `441` data-provider cases / `1,323` hermetic container executions, `99.4%` PCOV line coverage)

> [!NOTE] **Unofficial Engineering Showcase Notice (`drawlib-slim`)**: This diagnostic report is an independent, automated engineering and architecture assessment generated with **PNI501** and **Drawlib** to showcase **Slim Framework 4**'s standards-driven PHP design. It is part of the 4-portal **drawlib-slim** showcase:
>
> - **Portal 1 — User Guide**: https://slim.drawlib.com/user_guide/
> - **Portal 2 — API Reference Catalog**: https://slim.drawlib.com/api/
> - **Portal 3 — Internal Architecture Deep-Dive**: https://slim.drawlib.com/architecture/
> - **Portal 4 — PNI501 Engineering & Architecture Diagnostic**: https://slim.drawlib.com/pni501/

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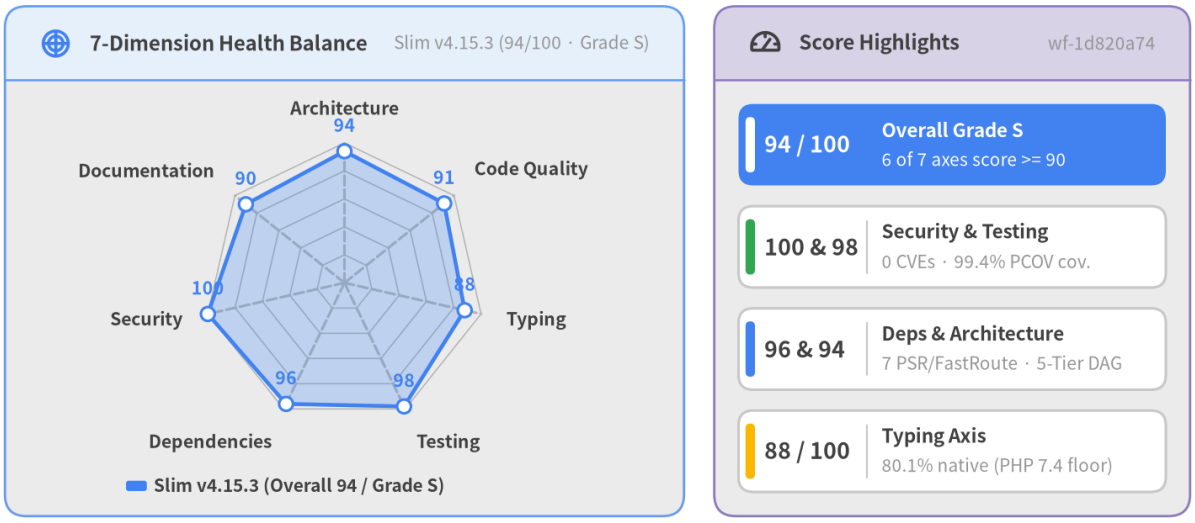
4. Audit Standards, Rule Compliance & Findings

5. Deep Dive: Code Quality, Testing & Runtime

6. Deep Dive: Security, Supply Chain & API Surface

1. Executive Summary

Slim Framework `v4.15.3` achieves an overall **94/100 (Grade S)** engineering and architecture health score (`12/17` evaluated rules passed, `3 WARN` , `0 FAIL` , `2 NOT_APPLICABLE`). Built around PHP-FIG **PSR-7**, **PSR-11**, **PSR-15**, and **PSR-17** standards alongside Nikic FastRoute (`nikic/fast-route ^1.3`), the **72-file (`3,149` LOC)** codebase maintains **100%** `declare(strict_types=1)` compliance across all source and test files, **98.6% PSR-12 import hygiene**, **99.4% measured PCOV line coverage** (`1,049 / 1,055` statements covered across `441` offline PHPUnit test cases / `1,323` hermetic container assertions), **zero CWE dangerous sinks**, and **zero Trivy supply-chain CVEs** across all 93 locked Composer packages.



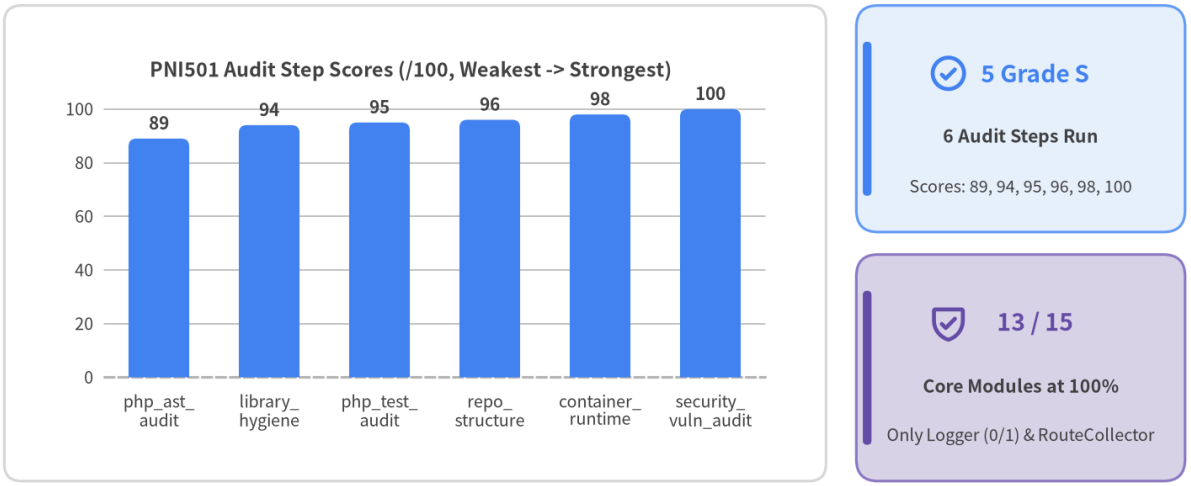
Slim Framework v4.15.3 — 7-Dimension Engineering Health Radar & Headline Axis Callouts

1.1 Seven-Dimension Engineering Scorecard

Six of the seven evaluation axes score `90/100` **or higher** (Grade `S`), led by a perfect **100/100** in Security and **98/100** in Testing:

Engineering Dimension	Axis Score	Grade	Key Evidence & Diagnostic Drivers
Security	`100 / 100`	**S**	`0` core or dev CVEs across `93` locked Composer packages; `0` secrets; `0` IaC misconfigurations; `0` CWE sinks; safe `htmlentities()` & XML entity disabling.
Testing	`98 / 100`	**S**	`99.4%` measured PCOV line coverage (`1,049 / 1,055` statements); `100%` hermetic `--network=none` pass rate; `AdrianSuter\Autoload\Override` SAPI mocks.
Dependencies	`96 / 100`	**S**	Ultra-lean 7-package runtime footprint (`6` official PHP-FIG PSR contract packages + `nikic/fast-route ^1.3`); dual-major PSR `^1.0 || ^2.0` SemVer ranges.
Architecture	`94 / 100`	**S**	Clean 5-Tier PSR-7 / PSR-15 micro-kernel and concentric LIFO middleware pipeline; 17 contract interfaces in `Slim\Interfaces\` ; only 1 localized 2-node cycle.
Code Quality	`91 / 100`	**S**	`100%` `declare(strict_types=1)` across `72` source files; `98.6%` PSR-12 import ordering; avg `43.7` LOC/file; `1` complexity hotspot (`MiddlewareDispatcher::addDeferred`).
Documentation	`90 / 100`	**S**	Complete repository governance (`README.md` , `CHANGELOG.md` , `CONTRIBUTING.md` , `SECURITY.md` , `CODE_OF_CONDUCT.md`); `0` PHPDoc signature mismatches; `71.6%` AST PHPDoc coverage.
Typing	`88 / 100`	**A**	`80.1%` fully typed callables at the native PHP AST level; `62` polymorphic `$middleware` / `$toResolve` parameters use PHPDoc unions (`callable|string|MiddlewareInterface`) for PHP `7.4` compatibility.

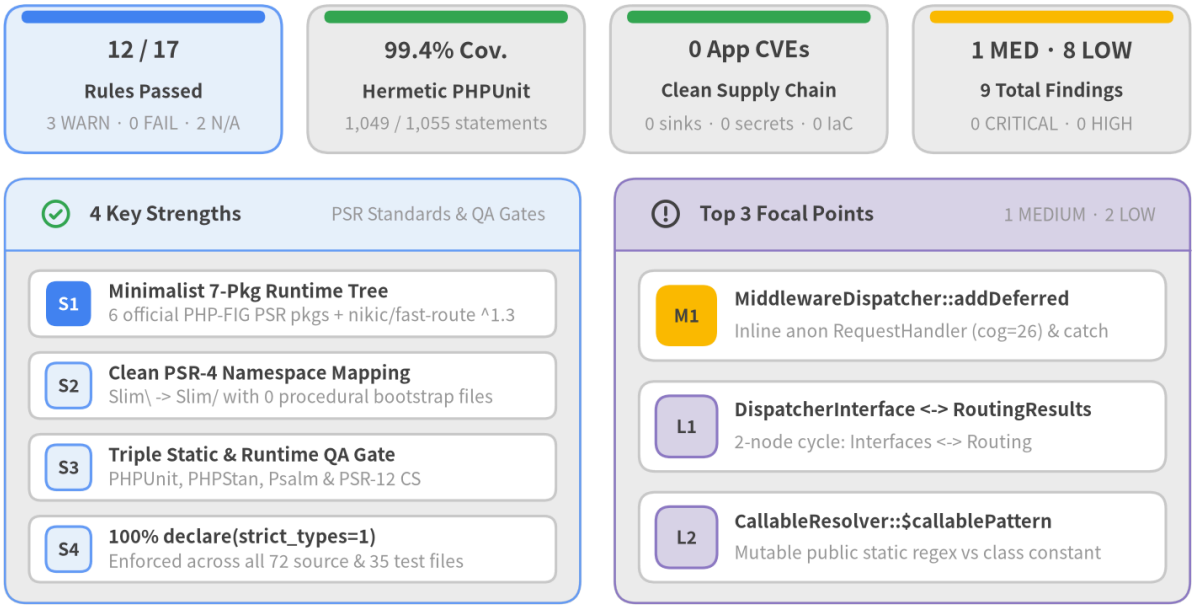
1.2 Audit Step Execution & Verification Summary



Slim Framework v4.15.3 — Audit Step Scores (Ascending: 89 to 100) & Verification Gates

Verification Pillar	Status / Score	Key Quantitative Evidence
Lowest-Scoring Audit Steps	`89` – `95`	`php_ast_audit=89 (A)`, `library_hygiene_audit=94 (S)`, `php_test_audit=95 (S)`, `repo_structure_audit=96 (S)`, `container_runtime_audit=98 (S)`, `security_vulnerability_audit=100 (S)`.
Container Runtime Verification	`passed` (`98/100`)	`php:8.2-cli-bookworm` + PCOV `1.0.12` : `build=passed` (`0.66s`), `php -l` passed across all `72` files, offline PHPUnit passed (`441` cases / `1,323` executions, `0` failures), `line_cov=99.4%`, `smoke=passed`.
Security & Supply-Chain Scan	`passed` (`100/100`)	`app_cves=0` (`CRITICAL=0`, `HIGH=0`, `MEDIUM=0`, `LOW=0`) across `7` core and `86` dev Composer packages; `misconfigs=0`; `secrets=0`; `ast_dangerous_sinks=0`.

1.3 Key Engineering Strengths vs. Top Architectural Risks



Slim Framework v4.15.3 — Headline Health KPIs, Key Engineering Strengths & Top Focal Points

Verified Engineering Strengths

1. **Minimalist, Zero-Bloat Runtime Dependency Tree**: `composer.json` restricts production dependencies to **6 official PHP-FIG PSR contract packages** (`psr/http-message`, `psr/http-factory`, `psr/http-server-handler`, `psr/http-server-middleware`, `psr/container`, `psr/log`) plus `nikic/fast-route (^1.3)`, with dual-major union ranges (`^1.0 || ^2.0`) that eliminate version conflicts in downstream applications.
2. **Clean PSR-4 Namespace Mapping**: Maps `'Slim\\' -> 'Slim/'` and `'Slim\\Tests\\' -> 'tests/'` with zero procedural bootstrap files (`files: []`) and zero import-time side effects.
3. **Triple Static & Runtime QA Gate**: `composer.json` configures PHPUnit (`^9.6 || ^10 || ^11 || ^12`), **PHPStan**, **Psalm**, and **PHP_CodeSniffer (PSR-12)** in unified verification scripts.
4. **100%** `declare(strict_types=1)` **Discipline**: Enforced at line 1 of all `72` production PHP files under `Slim/` and all test files under `tests/`.

Top Architectural & Code Quality Risks

Zero `CRITICAL` or `HIGH` findings exist across the repository. The single `[MEDIUM]` finding and top `[LOW]` advisories (mapped to concrete actions in **Chapter 3**) are:

1. `[MEDIUM]` `common/maintainability` — `MiddlewareDispatcher::addDeferred` (`Slim/MiddlewareDispatcher.php:144`): Spans `101` lines (`cognitive=26`) by embedding a `68`-line inline anonymous `RequestHandlerInterface` class (`cognitive=26`) that catches and ignores `RuntimeException` at line `186` when falling back from `CallableResolverInterface::resolve()` to custom PSR-15 resolution.
2. `[LOW]` `common/architecture` — `DispatcherInterface <-> RoutingResults` **2-Node Cycle**: `Slim\Interfaces\DispatcherInterface` imports and returns concrete `Slim\Routing\RoutingResults`, while `RoutingResults::__construct` imports `DispatcherInterface`, forming the sole cross-tier cycle between `Tier 1` and `Tier 3`.
3. `[LOW]` `php/code_quality` — `CallableResolver::$callablePattern` (`Slim/CallableResolver.php:34`): Declares `public static $callablePattern = '!^([^\:]+)\:([a-zA-Z_\x7f-\xff][a-zA-Z0-9_\x7f-\xff]*)$!'` as a mutable public static property rather than an immutable class constant (`public const CALLABLE_PATTERN`).

2. Architecture & Source Code Structure

Slim Framework `v4.15.3` organizes **72 PHP source files** (**3,149 LOC**, **72 classes/interfaces**, **311 callables**, average **43.7 LOC/file**) into a **5-Tier PSR-7 / PSR-15 Micro-Kernel & Concentric Middleware Pipeline** (**94/100**, Grade **S**). Every subsystem is decoupled through **17 explicit contract interfaces** in `Slim\Interfaces\`, enabling zero-configuration PSR-17 factory auto-discovery, LIFO PSR-15 middleware composition, Nikic FastRoute regex dispatching, and content-negotiated exception rendering.

Slim Framework v4.15.3 — Physical Namespace Layout & Subsystem Inventory (72 Files · 3,149 LOC)

Package Tree (Slim/) PSR-4

```
Slim/ (72 files · 3,149 LOC)
├── App.php (121 LOC)
├── MiddlewareDispatcher.php (177L)
├── CallableResolver.php (131 LOC)
├── ResponseEmitter & Logger (108L)
├── Factory/ (12 · 457 LOC)
│   └── AppFactory.php & Psr17/* (10)
├── Routing/ (11 · 913 LOC)
│   └── Route, RouteCollector, FastRoute*
├── Middleware/ (6 · 393 LOC)
├── Handlers/ & Error/ (10 · 469L)
├── Interfaces/ (17 · 229 LOC)
└── Exception/ (11 · 152 LOC)
```

Subsystem Inventory by Tier LOC & Core Classes

Module / Namespace	Tier	LOC	Key Types & Responsibility
App.php & Emitter	T5	215	App facade, PSR-15 handle & SAPI emit
Factory/* (2)	T5	221	AppFactory & ServerRequestCreator
MiddlewareDispatcher	T4	177	LIFO concentric PSR-15 onion kernel
Middleware/* (6)	T4	393	Routing, Error, BodyParsing, Buffer
Handlers/ & Error/	T4	386	ErrorHandler + HTML/JSON/XML render
Routing/* (11)	T3	913	RouteCollector, Route, FastRoute*
Handlers/Strategies	T3	83	RequestResponse, NamedArgs, Handler
CallableResolver.php	T2	145	PSR-11 DI & 'class:method' bridge
Factory/Psr17/* (10)	T2	236	Auto-discovery for 6 PSR-7 vendors
Interfaces & Exc (28)	T1	381	17 contracts & 11 HTTP exceptions

Slim Framework v4.15.3 — Source Directory Tree & Subsystem LOC Inventory (72 PHP Files, 3,149 LOC)

2.1 System Profile, PSR Dependency Stack & External Integrations

Slim Framework `v4.15.3` operates as a pure, standards-compliant **PHP library micro-framework** (`runtime_types = ["library"]` , PHP `~7.4.0 || ~8.0.0 .. ~8.5.0`) with

`declare(strict_types=1)` across 100% of source files. Instead of bundling a monolithic HTTP message implementation or proprietary dependency injection container, Slim depends strictly on **6 official PHP-FIG PSR interface packages** plus `nikic/fast-route`:

Runtime Dependency (`composer.json`)	Version Constraint	Architectural Role in Slim `v4.15.3`	Primary Consumer
`psr/http-message`	`^1.1 || ^2.0`	PSR-7 immutable `ServerRequestInterface`, `ResponseInterface`, `StreamInterface`, and `UriInterface`.	`App`, `ResponseEmit` `Middleware/*`, `Rou` `Handlers/*`
`psr/http-factory`	`^1.1`	PSR-17 `ResponseFactoryInterface` and `StreamFactoryInterface` for zero-coupling response synthesis.	`AppFactory`, `RouteCollector`, `R` `ErrorHandler`, `Psr`
`psr/http-server-handler`	`^1.0`	PSR-15 `RequestHandlerInterface` (`handle(ServerRequestInterface) : ResponseInterface`).	`App`, `MiddlewareDispatc` `RouteRunner`, `Rout` `Strategies/Reques` `r`
`psr/http-server-middleware`	`^1.0`	PSR-15 `MiddlewareInterface` (`process(ServerRequestInterface , RequestHandlerInterface)`).	`MiddlewareDispatc` `RoutingMiddleware` `ErrorMiddleware`, `BodyParsingMiddle`
`psr/container`	`^1.0 || ^2.0`	PSR-11 `ContainerInterface` (`has()`, `get()`) for resolving controllers, middlewares, and services.	`AppFactory`, `CallableResolver`, `MiddlewareDispatc` `RouteCollector`
`psr/log`	`^1.1 || ^2.0 || ^3.0`	PSR-3 `LoggerInterface` and `AbstractLogger` for structured exception reporting.	`ErrorMiddleware`, `ErrorHandler`, `Log`
`nikic/fast-route`	`^1.3`	High-speed regex route compilation (`RouteParser\Std`, `DataGenerator\GroupCountBased`, `Dispatcher\GroupCountBased`).	`Dispatcher`, `FastRouteDispatch` `RouteParser`

External Integration Boundaries

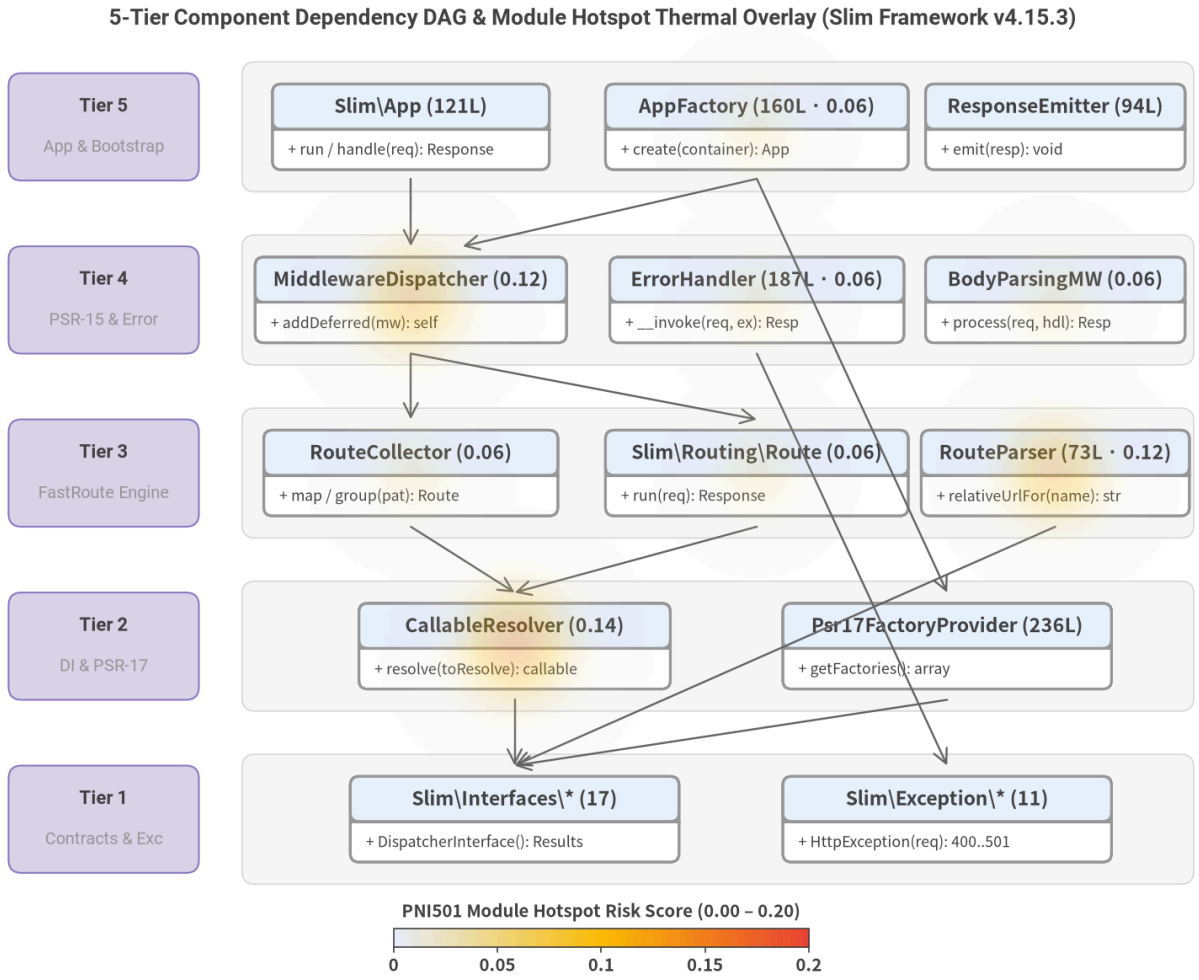
1. **PSR-17 / PSR-7 Auto-Discovery Providers** (`Slim\Factory\Psr17\Psr17FactoryProvider`): Automatically detects and wires any installed PSR-7/PSR-17 implementation in priority order: `Slim\Psr7` (`slim/psr7`), `Nyholm\Psr7` (`nyholm/psr7` + `nyholm/psr7-server`), `GuzzleHttp\Psr7` (`guzzlehttp/psr7`), `Laminas\Diactoros` (`laminas/laminas-diactoros`),

`HttpSoft\Message` (`httpsoft/http-message`), and optional `Slim\Http` (`slim/http`) message decorators.

2. **PSR-11 Dependency Injection Containers**: `AppFactory::createFromContainer($container)`, `CallableResolver`, and `MiddlewareDispatcher` resolve route callables, `'class:method'` notation, and invokable middleware services from PHP-DI, League Container, Symfony DependencyInjection, or Laminas ServiceManager.
3. **PHP SAPI & Long-Running Worker Runtimes**: `ResponseEmitter` streams status lines, multi-value HTTP headers (`Set-Cookie` safe), and chunked bodies to standard PHP-FPM/CLI SAPIs, while `App::handle(ServerRequestInterface)` supports stateful worker loops (RoadRunner, Swoole, FrankenPHP) without global state leakage.

2.2 Five-Tier PSR Architecture & Thermal Risk Score Overlay

Slim enforces a strict top-down dependency hierarchy across **five architectural tiers** (`Tier 5` down to `Tier 1`). Figure 2.2 overlays PNI501's pre-sorted `module_hotspots` risk scores (`CallableResolver.php=0.14`, `MiddlewareDispatcher.php=0.12`, `RouteParser.php=0.12`, and `0.06` across `Route.php`, `ErrorHandler.php`, `RouteCollector.php`, `AppFactory.php`, and `BodyParsingMiddleware.php`) directly onto the 5-Tier UML component dependency graph.



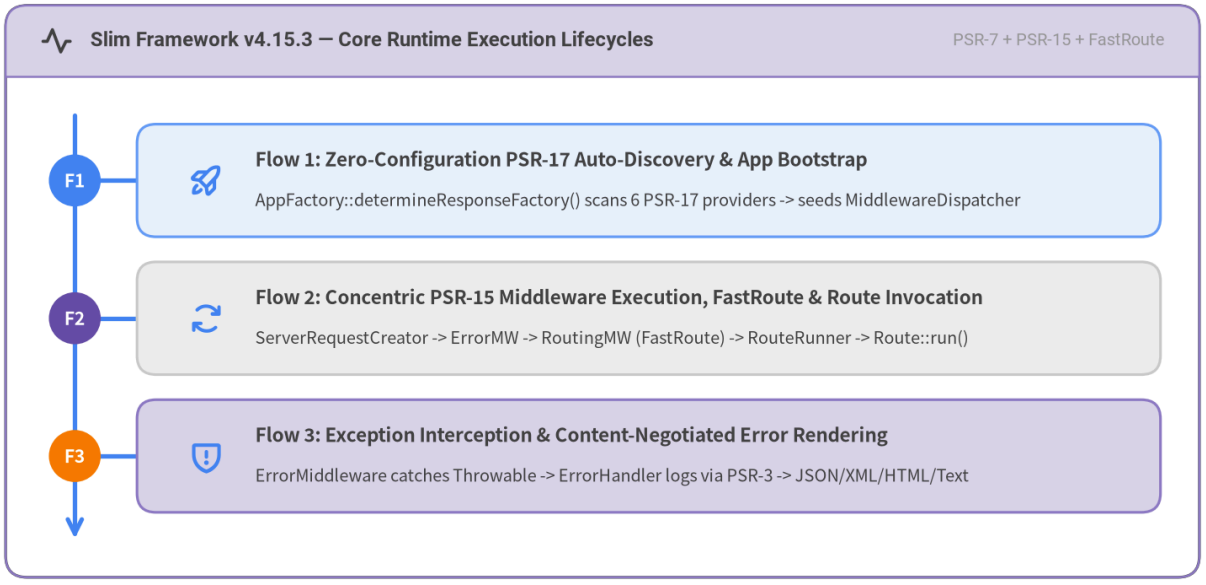
5-Tier PSR Component Dependency DAG & Module Hotspot Thermal Overlay (CallableResolver=0.14, MiddlewareDispatcher=0.12, RouteParser=0.12)

Five-Tier Boundary & Hotspot Analysis

Architectural Tier	Representative Source Paths	Allowed Dependencies	Boundary & Hotspot Assessment
Tier 5: Application Facade, Bootstrap Factories & SAPI Emitter	`Slim/App.php` (`121` LOC), `Slim/ResponseEmitter.php` (`94` LOC), `Slim/Factory/AppFactory.php` (`160` LOC, `risk=0.06`), `Slim/Factory/ServerRequestCreatorFactory.php` (`61` LOC)	`Tier 4`, `Tier 3`, `Tier 2`, `Tier 1`	**100% Clean**: `App` extends `RouteCollectorProxy` and implements `RequestHandlerInterface`, delegating cleanly downward.

Architectural Tier	Representative Source Paths	Allowed Dependencies	Boundary & Hotspot Assessment
Tier 4: PSR-15 Middleware Stack & Content-Negotiated Error Subsystem	`Slim/MiddlewareDispatcher.php` (`177` LOC, `risk=0.12`), `Slim/Middleware/*` (`6` files, `393` LOC; `BodyParsingMiddleware` `risk=0.06`), `Slim/Handlers/ErrorHandler.php` (`187` LOC, `risk=0.06`), `Slim/Error/Renderers/*` (`5` files, `199` LOC)	`Tier 3` , `Tier 2` , `Tier 1`	**Hotspot** `0.12` : `MiddlewareDispatcher::addDeferred` (`cognitive=26`) wraps deferred string/callable middlewares in an inline anonymous `RequestHandlerInterface` .
Tier 3: FastRoute Routing Engine & Handler Invocation Strategies	`Slim/Routing/Route.php` (`192` LOC, `risk=0.06`), `RouteCollector.php` (`174` LOC, `risk=0.06`), `RouteCollectorProxy.php` (`103` LOC), `RouteParser.php` (`73` LOC, `risk=0.12`), `FastRouteDispatcher.php` (`64` LOC), `RoutingResults.php` (`59` LOC), `Strategies/*` (`4` files, `83` LOC)	`Tier 2` , `Tier 1`	**Hotspot** `0.12` : `RouteParser::relativeUrlFor` (`cognitive=13`) computes relative path segments; `Route::__construct` accepts `9` collaborators.
Tier 2: PSR-17 Provider Adapters & Container Callable Resolver	`Slim/CallableResolver.php` (`131` LOC, `risk=0.14`), `Slim/Logger.php` (`14` LOC), `Slim/Factory/Psr17/*` (`10` files, `236` LOC)	`Tier 1`	**Hotspot** `0.14` : `CallableResolver` bridges PSR-11 containers and `'class:method'` strings (`$callablePattern` static regex + PHP 7.4 untyped `$toResolve`).
Tier 1: Core Interfaces & HTTP Specialized Exception Hierarchy	`Slim/Interfaces/*` (`17` files, `229` LOC), `Slim/Exception/*` (`11` files, `152` LOC: `400` , `401` , `403` , `404` , `405` , `410` , `429` , `500` , `501`)	`[]` *(Leaf Tier)*	**1 Cross-Layer Cycle**: `Slim\Interfaces\DispatcherInterface` imports concrete `Slim\Routing\RoutingResults` (`Tier 3`).

2.3 Core Runtime Execution Flows & Concurrency Model



Slim Framework v4.15.3 — Three Core Runtime Execution Pipelines (Bootstrap, LIFO PSR-15 Dispatch, Exception Rendering)

Detailed Execution Flow Walkthrough

1. **Flow 1 — Zero-Configuration PSR-17 Auto-Discovery & App Bootstrap** (`AppFactory::create` / `createFromContainer`):
   - `AppFactory::determineResponseFactory()` iterates through `Psr17FactoryProvider::$factories` (`SlimPsr17Factory`, `NyholmPsr17Factory`, `GuzzlePsr17Factory`, `LaminasDiactorosPsr17Factory`, `HttpSoftPsr17Factory`) until it locates an installed PSR-17 `ResponseFactoryInterface`.
   - `AppFactory` instantiates `CallableResolver`, `RouteCollector` (defaulting to the `RequestResponse` invocation strategy), `RouteResolver`, `RouteRunner`, and `MiddlewareDispatcher` seeded with `RouteRunner` as its innermost kernel.
   - Application bootstrap code registers routes (`$app->get()`, `$app->post()`, `$app->group()`) on `RouteCollectorProxy` and pushes PSR-15 middlewares (`$app->add()`, `$app->addBodyParsingMiddleware()`, `$app->addRoutingMiddleware()`, `$app->addErrorMiddleware()`) onto `MiddlewareDispatcher`'s LIFO stack.

2. **Flow 2 — Concentric PSR-15 Middleware Execution, FastRoute Matching & Route Invocation (`App::run` / `App::handle`):**
    - When `$request` is `null` in `App::run()`, `ServerRequestCreatorFactory::create()->createServerRequestFromGlobals()` synthesizes an immutable PSR-7 `ServerRequestInterface` from PHP superglobals.
    - `MiddlewareDispatcher::handle($request)` traverses the LIFO middleware onion from outermost (typically `ErrorMiddleware`) inward through `RoutingMiddleware` and `BodyParsingMiddleware`.
    - `RoutingMiddleware::performRouting()` invokes `RouteResolver::computeRoutingResults()` -> `Dispatcher` -> `FastRouteDispatcher`, attaching `RouteContext::ROUTE`, `ROUTE_PARSER`, and `ROUTING_RESULTS` attributes to `$request` (or throwing `HttpNotFoundException` / `HttpMethodNotAllowedException`).
    - At the center of the application middleware stack, `RouteRunner::handle()` resolves `RouteGroup` middlewares and delegates to `Route::run()`, which executes route-specific middlewares before invoking the matched callable via `InvocationStrategyInterface` (`RequestResponse`, `RequestResponseArgs`, `RequestResponseNamedArgs`, or `RequestHandler`).
    - Finally, `ResponseEmitter::emit($response)` writes the HTTP status line, headers, and chunked body stream (`responseChunkSize = 4096`) to the PHP SAPI output buffer.
3. **Flow 3 — Exception Interception & Content-Negotiated Error Rendering (`ErrorMiddleware::process`):**
    - `ErrorMiddleware::process()` catches any `Throwable` inside its `try/catch` boundary and delegates to `ErrorMiddleware::handleException($request, $exception)`.
    - `ErrorMiddleware` looks up a registered exact-match or subclass-match handler in `$this->handlers`, falling back to the default `ErrorHandler`.
    - `ErrorHandler::__invoke()` determines the HTTP status code (`400..501` from `HttpException` or `500` for generic `Throwable`), logs the error via PSR-3 `LoggerInterface` when `$logErrors` is enabled, and inspects the request `Accept` header via `determineRenderer()` to serialize the payload through `JsonErrorRenderer`, `XmlErrorRenderer`, `HtmlErrorRenderer`, or `PlainTextErrorRenderer` with CORS and `Allow` headers preserved.

3. Recommended Remediation Roadmap

Slim Framework `v4.15.3` enters remediation planning from an exceptional **94/100 (Grade S)** baseline with **zero** `P0` **urgent blockers**, **zero** `FAIL` **compliance rules**, **zero** `CRITICAL` **or** `HIGH` **source-code findings**, and **zero supply-chain CVEs** across all 93 locked Composer packages. Rather than emergency defect patching, the 7-item engineering roadmap focuses on **1** `P1` **high-value maintainability refactoring** in `MiddlewareDispatcher::addDeferred` followed by **6** `P2` **architectural boundary, documentation, testing, and Slim 5.x typing modernization items**.

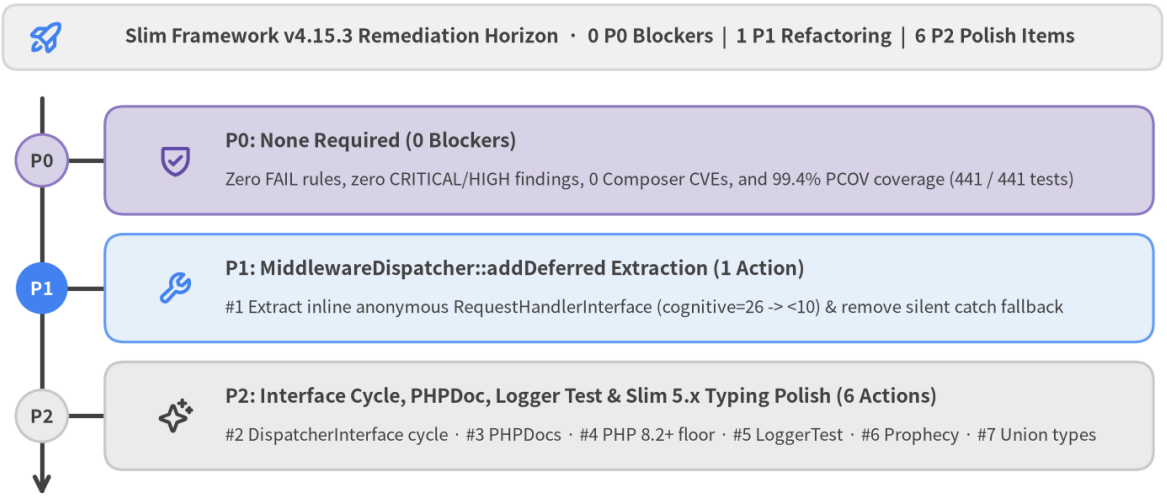


Figure 3.1 — Phased Engineering Refinement Roadmap for Slim Framework v4.15.3 (0 P0 Blockers · 1 P1 High-Leverage · 6 P2 Polish)

3.1 Action Plan Traceability Matrix (`P0 -> P1 -> P2`)

Every remediation candidate identified by PNI501 maps deterministically to concrete file coordinates, expected engineering outcomes, and underlying compliance rules:

Remediation Action Plan Traceability Matrix (0 P0 Urgent · 1 P1 High-Leverage · 6 P2 Polish)

Priority	Action Title	Target Files	Expected Impact	Related Rules
[P1] #1 maintainability	Refactor MiddlewareDispatcher::addDeferred inline anonymous handler into helper class	Slim/MiddlewareDispatcher.php (lines 144–244, line 186)	Cuts cognitive complexity 26 -> <10 & removes empty catch(RuntimeException)	common/maintainability php/code_quality
[P2] #2 architecture	Decouple DispatcherInterface from concrete Slim\Routing\RoutingResults	Slim/Interfaces/DispatcherInterface.php Slim/Routing/RoutingResults.php	Eliminates the sole 2-node circular namespace dependency in the codebase	common/architecture
[P2] #3 documentation	Add @inheritdoc or PHPDoc summaries to ErrorRenderer & HttpException public methods	Slim/Error/Renderers/* Slim/Exception/HttpException Slim/Factory/AppFactory.php	Increases AST PHPDoc coverage from 71.6% to >90% with 0 mismatches	common/documentation
[P2] #4 modernization	Drop EOL PHP < 8.2 floor in next major version branch (Slim 5.x)	composer.json Slim/* (Slim 5.x branch)	Unlocks readonly properties, constructor promotion & union types across Slim core	php/modern_stack
[P2] #5 testing	Add unit test for Slim\Logger::log() fallback PSR-3 logger	tests/LoggerTest.php Slim/Logger.php:30	Raises measured PCOV line coverage from 99.4% to 99.5%+ (14/15 modules 100%)	common/runtime_verification
[P2] #6 testing	Pair Prophecy spy predictions in AppTest & ErrorMiddlewareTest with explicit assertions	tests/AppTest.php tests/Middleware/ErrorMiddlewareTest.php	Eliminates reliance on implicit Prophecy teardown checks alone (20 tests)	php/testing
[P2] #7 typing	Add native union type hints & const CALLABLE_PATTERN in CallableResolver & Dispatcher	Slim/CallableResolver.php Slim/MiddlewareDispatcher.php Slim/Routing/Route*.php	Raises native AST type safety from 80.1% toward 100% & removes static state	php/typing php/code_quality

Figure 3.2 — Action Plan Traceability Matrix Mapping Priorities, Target Files, Expected Impact, and Audit Rules

---

3.2 Detailed Engineering Specifications by Action Item

1. `[P1]` Refactor `MiddlewareDispatcher::addDeferred` Inline Anonymous Handler (`maintainability`)

- **Target Coordinates**: `Slim/MiddlewareDispatcher.php:144–244` (`MiddlewareDispatcher::addDeferred` and its nested anonymous `RequestHandlerInterface::handle` at lines `173–240` , plus the empty `catch (RuntimeException $e)` block at line `186`).
- **Diagnostic Root Cause**: `addDeferred()` wraps string or callable middleware definitions inside a `68` -line inline anonymous `RequestHandlerInterface` class (`cognitive=26`). When `$this->callableResolver` is not an `AdvancedCallableResolverInterface` , it

attempts `$this->callableResolver->resolve($this->middleware)` inside a `try { ... } catch (RuntimeException $e) {}` block and silently ignores resolution failures before falling back to manual container/class instantiation.

- **Recommended Implementation**:
  1. Extract the deferred resolution handler into a dedicated internal class (`Slim\DeferredMiddlewareHandler implements RequestHandlerInterface`) or decompose resolution into a private `resolveDeferredMiddleware()` helper method.
  2. Delegate directly to `AdvancedCallableResolverInterface::resolveMiddleware()` when available, and replace the empty `catch (RuntimeException)` block with explicit container/class existence checks so genuine instantiation errors are never swallowed.
- **Expected Impact**: Reduces cognitive complexity in `Slim/MiddlewareDispatcher.php` from `26` to `<10` and eliminates the sole `bug_risk_idioms` finding.

2. `[P2]` Decouple `Slim\Interfaces\DispatcherInterface` from Concrete `Slim\Routing\RoutingResults` (`architecture`)

- **Target Coordinates**: `Slim/Interfaces/DispatcherInterface.php:10,27` and `Slim/Routing/RoutingResults.php:13,47`.
- **Diagnostic Root Cause**: `Slim\Interfaces\DispatcherInterface::dispatch(string $httpMethod, string $uri): RoutingResults` imports and returns concrete `Slim\Routing\RoutingResults` (`Tier 3`), while `RoutingResults::__construct(DispatcherInterface $dispatcher, ...)` imports `DispatcherInterface` (`Tier 1`). Additionally, `Slim\Interfaces\RouteGroupInterface` imports concrete `Slim\MiddlewareDispatcher`, and `Slim\Interfaces\RouteResolverInterface` imports `RoutingResults`.
- **Recommended Implementation**: Introduce `Slim\Interfaces\RoutingResultsInterface` (carrying `NOT_FOUND = 0`, `FOUND = 1`, `METHOD_NOT_ALLOWED = 2` and accessor contracts) so `Slim\Interfaces\` remains a 100% leaf-level contract package with zero upward imports into `Slim\Routing\`.

3. [P2] Add @inheritdoc or PHPDoc Summaries to ErrorRenderer and HttpException Methods (documentation)

- **Target Coordinates**: `Slim/Error/Renderers/HtmlErrorRenderer.php`, `JsonErrorRenderer.php`, `PlainTextErrorRenderer.php`, `XmlErrorRenderer.php`, `Slim/Exception/HttpException.php`, and setter methods on `Slim/Factory/AppFactory.php`.
- **Diagnostic Root Cause**: While `doc_signature_mismatches = 0` and `empty_docstrings = 0`, `98` inherited or self-explanatory accessor methods omit explicit `/** ... */` docblocks, resulting in `71.6%` AST PHPDoc coverage.
- **Recommended Implementation**: Add concise `/** @inheritdoc */` annotations on `ErrorRendererInterface::__invoke` implementations and one-line summaries on `HttpException` getters/setters and `AppFactory` static setters, lifting AST PHPDoc coverage above `90%`.

4. [P2] Drop EOL PHP < 8.2 Compatibility Floor in Slim 5.x (modernization)

- **Target Coordinates**: `composer.json` (`require.php: "~7.4.0 || ~8.0.0 || ~8.1.0 || ~8.2.0 || ~8.3.0 || ~8.4.0 || ~8.5.0"`).
- **Recommended Implementation**: Preserve the PHP `7.4+` compatibility floor on the `4.x` maintenance branch for enterprise LTS consumers, while setting `"php": "^8.2"` in the `5.x` branch to unlock `readonly` properties, constructor property promotion, and native union/intersection types.

5. [P2] Add Unit Test for Slim\Logger::log() Fallback PSR-3 Logger (testing)

- **Target Coordinates**: `Slim/Logger.php:30` (`error_log(sprintf('%s: %s', $level, $message), 4);`) and `tests/LoggerTest.php`.
- **Diagnostic Root Cause**: PCOV Clover coverage reaches `99.4%` (`1,049 / 1,055` statements), with `Slim\Logger::log()` (`0/1` statements) and `5` defensive lines in `Slim\Routing\RouteCollector` (`69/74` statements, `93.2%`) accounting for all `6` uncovered lines in the entire framework.

- **Recommended Implementation**: Add `tests/LoggerTest.php` using `AdrianSuter\Autoload\Override` (already configured in `tests/bootstrap.php`) to intercept `Slim\error_log` and assert the formatted `"{$level}: {$message}"` payload and `message_type = 4` (`SAPI` log handler).

6. `[P2]` Pair Prophecy Spy Predictions with Explicit State Assertions (`testing`)

- **Target Coordinates**: `20` test methods across `tests/AppTest.php`, `tests/Middleware/ErrorMiddlewareTest.php`, `tests/Handlers/ErrorHandlerTest.php`, `tests/ResponseEmitterTest.php`, and `tests/Routing/RouteResolverTest.php`.
- **Recommended Implementation**: Pair `$prophecy->shouldBeCalled()` spy expectations with explicit `$this->assertSame()` / `$this->assertInstanceOf()` assertions on the returned `ResponseInterface` or `RoutingResults` object.

7. `[P2]` Adopt Native Union Types & `const CALLABLE_PATTERN` (`typing` / `code_quality`)

- **Target Coordinates**: `Slim/CallableResolver.php:34` (`public static $callablePattern`), plus `62` polymorphic methods across `Slim/CallableResolver.php`, `Slim/MiddlewareDispatcher.php`, `Slim/App.php`, and `Slim/Routing/RouteCollectorProxy.php`, and `Slim/Routing/Route.php:120` (`9`-parameter constructor).
- **Recommended Implementation**:
  1. Convert `CallableResolver::$callablePattern` to `public const CALLABLE_PATTERN = '!^([^\:]+)\:([a-zA-Z_\x7f-\xff][a-zA-Z0-9_\x7f-\xff]*)$!';` (retaining a deprecated static property alias if needed in `4.x`).
  2. Replace PHPDoc-only `@param MiddlewareInterface|string|callable $middleware` annotations with native PHP 8.0+ union types (`MiddlewareInterface|string|callable` and `callable|array|string`) in `Slim 5.x`, and encapsulate `Route::__construct`'s 9 parameters using a `RouteConfiguration` value object.

4. Audit Standards, Rule Compliance & Findings

PNI501 evaluated Slim Framework `v4.15.3` across **6 deterministic audit steps** and **17 engineering compliance rules** spanning static PHP AST maintainability, PSR-12 import ordering, Composer supply-chain hygiene, PSR library interface design, PHPUnit test determinism, and hermetic `--network=none` container execution. Every audit step achieved Grade `A` or `S` (`89/100` to `100/100`), yielding **12** `PASS` **rules**, **3 advisory** `WARN` **rules**, **0** `FAIL` **rules**, **2** `NOT_APPLICABLE` **framework rules** (`laravel`, `wordpress`), and **9 consolidated location findings** (`0 CRITICAL`, `0 HIGH`, `1 MEDIUM`, `8 LOW`).



Figure 4.1 — Executed PNI501 Audit Step Scores (Weakest-First Ascending: 89 to 100) & 17-Rule Stacked Compliance Breakdown

4.1 Executed Audit Step Breakdown (`Weakest -> Strongest`)

Audit Step ID	Score & Grade	Rule Verdicts	Step Synthesis & Key Evidence
`php_ast_audit`	`89 / 100` **(A)**	`5 PASS`, `2 WARN`, `0 FAIL`, `2 N/A`	`72` source files (`3,149` LOC, `72` classes/interfaces, `311` callables); `100%` `declare(strict_types=1)`, `98.6%` PSR-12 import ordering (`1` out-of-order `use` in `RequestResponseNamedArgs.php:16`), `0` CWE sinks, `0` direct superglobal reads in routing/middleware. Advisories: `MiddlewareDispatcher::addDeferred` (`cognitive=26`), `DispatcherInterface <-> RoutingResults` 2-node cycle, `Route::__construct` (`9` params), and `62` PHP 7.4 polymorphic untyped parameters.
`library_hygiene_audit`	`94 / 100` **(S)**	`2 PASS`, `1 WARN`, `0 FAIL`	Exposes `17` explicit interfaces in `Slim\Interfaces\`, `0` import-time side effects, `0` `exit` / `die` statements, flexible SemVer `^` and `||` bounds across all `7` runtime packages (`psr/container: ^1.0 || ^2.0`, `psr/http-message: ^1.1 || ^2.0`, `psr/log: ^1.1 || ^2.0 || ^3.0`), `0` scattered `getenv()` / `$_ENV` reads, and `71.6%` AST PHPDoc coverage (`0` signature mismatches).
`php_test_audit`	`95 / 100` **(S)**	`1 PASS`, `0 WARN`, `0 FAIL`	`35` PHPUnit test classes and `306` test methods under `tests/` mirroring `Slim/`. Runs 100% offline using in-memory PSR-7 streams (`PSR7ObjectProvider`), `AdrianSuter\Autoload\Override` for deterministic `header()` / `connection_status()` / `is_writable()` interception, and `executionOrder="random"` in `phpunit.xml.dist`.
`repo_structure_audit`	`96 / 100` **(S)**	`6 PASS`, `0 WARN`, `0 FAIL`	Exemplary Composer packaging (`composer.json` + deterministic `composer.lock` with `93` locked packages), `0` unpinned dependencies, triple static analysis (`PHPStan`, `Psalm`, `PHP_CodeSniffer` PSR-12), and full governance (`README.md`, `CHANGELOG.md`, `CONTRIBUTING.md`, `CODE_OF_CONDUCT.md`, `SECURITY.md`, `LICENSE-SLIM`).
`container_runtime_audit`	`98 / 100` **(S)**	`2 PASS`, `0 WARN`, `0 FAIL`	Hermetic container verification (`php:8.2-cli-bookworm` + Composer 2 + PCOV `1.0.12`) passed all 4 gates (`RV-01` .. `RV-04`): image build (`0.66s`), `php -l` syntax check across `72` files, `--network=none` PHPUnit 11.5.57 (`441` cases / `1,323` executions, `0` failures), **99.4% PCOV line coverage** (`1,049 / 1,055` statements; `13/15` core modules at `100.0%`), and PSR-4 autoloader smoke check.
`security_vulnerability_audit`	`100 / 100` **(S)**	`3 PASS`, `0 WARN`, `0 FAIL`	Trivy `0.69.3` SCA, secret, and IaC scanning + PHP SAST confirm `0` **CVEs** across all `93` locked Composer packages (`core_app_vuln_counts=0`, `dev_app_vuln_counts=0`), `0` **secrets**, `0` **misconfigurations**, and `0` **CWE dangerous sinks**.

4.2 Full 17-Rule Compliance Matrix (`FAIL -> WARN -> PASS -> NOT_APPLICABLE`)

Across all 17 evaluated PNI501 engineering standards, Slim Framework `v4.15.3` records `0 FAIL`, `3 WARN`, `12 PASS`, and `2 NOT_APPLICABLE` checks:

Status	Rule ID	Evaluated By	Diagnostic Evidence & Verification Summary
`[WARN]`	`common/documentation`	`library_hygiene_audit`	`README.md` (`101` lines), `CHANGELOG.md`, `CONTRIBUTING.md`, `CODE_OF_CONDUCT.md`, and `SECURITY.md` provide complete governance; `0` PHPDoc `@param` signature mismatches (`doc_signature_mismatches=0`) and `0` empty docstrings; `71.6%` overall AST PHPDoc coverage (`98` inherited/accessor methods omit explicit `/** ... */` blocks).
`[WARN]`	`common/maintainability`	`php_ast_audit`	`3,149` LOC across `72` files (avg `43.7` LOC/file), with `3` complex methods (`MiddlewareDispatcher::addDeferred` `cognitive=26`, inline anonymous `RequestHandlerInterface::handle` `cognitive=26`, `RouteParser::relativeUrlFor` `cognitive=13`) and `1` constructor exceeding 6 parameters (`Slim\Routing\Route::__construct`, `params=9`).
`[WARN]`	`php/typing`	`php_ast_audit`	`100%` of files declare `declare(strict_types=1)` and `80.1%` of callables are fully typed at the native PHP AST level; `62` callables omit native parameter type hints on polymorphic `$middleware` / `$toResolve` / `$callable` arguments (`callable|string|MiddlewareInterface`) due to PHP `7.4` union-type constraints, plus `6` unscoped `@phpstan` / `@psalm` annotations.
`[PASS]`	`common/architecture`	`php_ast_audit`	Clean 5-Tier PSR subsystem layering (`Interfaces/Exception` -> `CallableResolver/Psr17` -> `Routing/Strategies` -> `Middleware/Handlers/Error` -> `App/Factory/ResponseEmitter`) with only `1` localized bidirectional reference between `Slim\Interfaces\DispatcherInterface` and `Slim\Routing\RoutingResults`.
`[PASS]`	`common/git_workflow`	`repo_structure_audit`	`.gitignore`, `.gitattributes`, and `.editorconfig` enforce clean working-tree hygiene and consistent LF line endings.
`[PASS]`	`common/library_guidelines`	`library_hygiene_audit`	`17` contract interfaces in `Slim\Interfaces\`, `0` import-time side effects, `0` `exit` / `die` calls, `0` `scattered_env_access`, and flexible `^1.0 || ^2.0` SemVer ranges across PSR packages.
`[PASS]`	`common/production_readiness`	`container_runtime_audit`	`0` external I/O calls without timeouts (`missing_io_timeouts=0`), structured PSR-3 `LoggerInterface` error propagation in `ErrorHandler`, and stateless PSR-15 `RequestHandlerInterface` execution compatible with PHP-FPM, RoadRunner, Swoole, and FrankenPHP.
`[PASS]`	`common/runtime_verification`	`container_runtime_audit`	`RV-01` harness build passed (`exit_code=0`, `0.66s`); `RV-02` offline `--network=none` PHPUnit suite passed (`0` failed, `0` errors); `RV-03` measured **99.4% line coverage** via `phpunit-pcov` (`1,049 / 1,055` statements); `RV-04` Composer PSR-4 autoloader smoke passed.

Status	Rule ID	Evaluated By	Diagnostic Evidence & Verification Summary
`[PASS]`	`common/security`	`repo_structure_audit`	`0` committed secret files (`committed_secret_files=[]`), explicit `SECURITY.md` disclosure policy, `0` hardcoded secrets (`hardcoded_secrets=0`), `0` dangerous sinks (`dangerous_sinks=0`), strict `htmlentities(..., ENT_COMPAT | ENT_HTML5, 'utf-8')` escaping in `HtmlErrorRenderer`, and external XML entity disabling in `BodyParsingMiddleware`.
`[PASS]`	`common/security_vulnerability`	`security_vulnerability_audit`	`0` CWE injection or deserialization sinks across all `72` PHP source files; `0` **CVEs** across all runtime and dev Composer dependencies (`core_app_vuln_counts=0`, `dev_app_vuln_counts=0`), `0` secrets, and `0` container IaC misconfigurations.
`[PASS]`	`common/supply_chain`	`repo_structure_audit`	`composer.json` and `composer.lock` (`93` locked packages) present with `0` unpinned dependencies; runtime dependencies limited strictly to `6` PSR contract packages and `nikic/fast-route ^1.3`.
`[PASS]`	`php/code_quality`	`php_ast_audit`	`98.6%` PSR-12 import hygiene (`1` out-of-order `use RuntimeException` in `Slim/Handlers/Strategies/RequestResponseNamedArgs.php:16`), `0` direct superglobal reads in core routing/middleware, `1` mutable static property (`CallableResolver::$callablePattern`), and `1` fallback catch block (`MiddlewareDispatcher.php:186`).
`[PASS]`	`php/dependencies`	`repo_structure_audit`	Clean PSR-4 autoload mapping (`'Slim\\': 'Slim/'`, `'Slim\\Tests\\': 'tests/'`) with `0` procedural file autoloads and dual static analyzers (`phpstan/phpstan: ^1 || ^2` and `vimeo/psalm: ^5 || ^6`).
`[PASS]`	`php/modern_stack`	`repo_structure_audit`	Supports active PHP `8.2`, `8.3`, `8.4`, and `8.5` runtimes alongside backward compatibility constraints (`~7.4.0` .. `~8.5.0`); `0` deprecated PHP functions; PSR-3 `LoggerInterface` used throughout `ErrorMiddleware` and `ErrorHandler`.
`[PASS]`	`php/testing`	`php_test_audit`	`306` PHPUnit test methods across `35` test files with `79.2%` direct AST symbol mapping (remaining files are declarative interfaces and marker HTTP exceptions) and 100% offline execution via in-memory PSR-7 streams and `AdrianSuter\Autoload\Override`.
`[NOT_APPLICABLE]`	`php/frameworks/laravel`	`php_ast_audit`	Repository is the upstream Slim Framework library, not a Laravel application.
`[NOT_APPLICABLE]`	`php/frameworks/wordpress`	`php_ast_audit`	Repository is the upstream Slim Framework library, not a WordPress plugin or theme.

4.3 Consolidated Location Findings Index (`CRITICAL -> INFO`)

Consolidated Location Findings Matrix (9 Unique Findings · 0 CRITICAL · 0 HIGH · 1 MEDIUM · 8 LOW)

Severity	Location / Scope	Rule ID	Diagnostic Evidence	Recommended Remediation
[MEDIUM] #1	Slim/Middleware Dispatcher.php:144,186	common/ maintainability	addDeferred (101L, cog=26) embeds anon RequestHandler (68L, cog=26) & empty catch(RuntimeException) at L186	Extract DeferredMiddlewareHandler class & prefer AdvancedCallable ResolverInterface::resolveMiddleware
[LOW] #2	Slim/Interfaces/ DispatcherInterface.php	common/ architecture	DispatcherInterface imports concrete RoutingResults while RoutingResults imports DispatcherInterface (2-node)	Extract RoutingResultsInterface in Slim\Interfaces\ so contracts have zero imports from Slim\Routing\
[LOW] #3	Slim/Error/Renderers/* Slim/Exception/*	common/ documentation	Public ErrorRenderer::__invoke and HttpException getters/setters omit explicit PHPDocs (71.6% coverage)	Add @inheritdoc or concise PHPDoc summaries on ErrorRenderer and HttpException public methods
[LOW] #4	Slim/Logger.php:30 RouteCollector.php	common/runtime_ verification	PCOV reports 99.4% line coverage, leaving Logger::log() L30 (0/1) & 5 RouteCollector lines uncovered	Add tests/LoggerTest.php exercising Slim\Logger::log() with an overridden error_log() SAPI stub
[LOW] #5	Slim/Callable Resolver.php:34	php/ code_quality	Declares public static callablePattern as a mutable public static property rather than a class constant	Convert to public const CALLABLE_PATTERN and allow pattern customization via constructor
[LOW] #6	composer.json (require.php)	php/ modern_stack	Retains '~7.4.0 || ~8.0.0 || ~8.1.0' for 4.x ecosystem compatibility, precluding readonly/union syntax	Keep 4.x floor for existing users while targeting PHP ^8.2 minimum in the Slim 5.x major release branch
[LOW] #7	Slim/CallableResolver MiddlewareDispatcher	php/ typing	62 methods omit native parameter types on middleware / toResolve due to PHP 7.4 union-type limits	Adopt native PHP 8.0+ union types (MiddlewareInterface|callable|string) once PHP 7.4 support is dropped
[LOW] #8	tests/AppTest.php ErrorMiddlewareTest	php/ testing	20 test methods verify behavior exclusively via Prophecy spy calls without explicit assert*() calls	Pair Prophecy shouldBeCalled() predictions with explicit return or state assertions where applicable
[LOW] #9	Slim/Routing/ Route.php:120	common/ maintainability	Route::__construct accepts 9 positional parameters, exceeding the SIG 6-parameter interfacing threshold	Group shared runtime collaborators into an immutable RouteConfiguration value object in Slim 5.x

Figure 4.2 — Consolidated Location Findings Overview (9 Unique Locations: 0 CRITICAL · 0 HIGH · 1 MEDIUM · 8 LOW)

4.4 Configured Scope & Accepted Design Waivers (`.pni/pni.toml`)

PNI501 inspected `.pni/pni.toml` for project-scoped rule waivers (`[[opt_out]]`). Exactly **1 design waiver** is configured for the ephemeral batch container test harness:

Waiver Rule ID	Scoped Glob Pattern	Matched Findings	Engineering Justification
`DS-0026`	`[.pni/dockerfiles/**]`	`0`	Hermetic batch test container (`php:8.2-cli-bookworm`) executes `phpunit` and exits immediately upon completion; long-running daemon `HEALTHCHECK` directives are not applicable.

5. Deep Dive: Code Quality, Testing & Runtime

This chapter examines Slim Framework `v4.15.3` across two complementary verification planes: **deterministic PHP AST static analysis** (`72` source files, `3,149` LOC, `72` classes/interfaces, `311` callables; `php_ast_audit` : **89/100, Grade A**) and **air-gapped container runtime execution** on `php:8.2-cli-bookworm` + PCOV `1.0.12` (`container_runtime_audit` : **98/100, Grade S**; `php_test_audit` : **95/100, Grade S**).

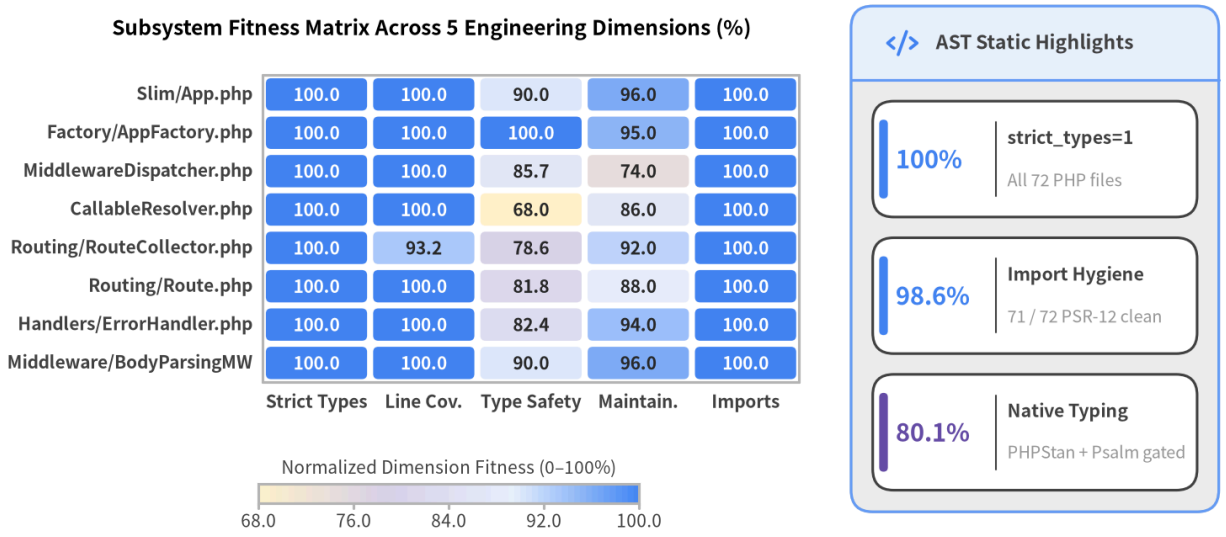


Figure 5.1 — Core Slim Subsystem Fitness Matrix Across 5 Engineering Dimensions & Static AST Highlights

5.1 PHP AST Static Quality, Typing & SIG Maintainability Analysis

Slim Framework `v4.15.3` is remarkably compact: its `72` production PHP files total just `3,149` **lines of code** (an average of `43.7` **LOC per file**) across `72` classes/interfaces and `311` callables.

Static Quality & SIG Metric	Measured Value	Status	Engineering Interpretation
`declare(strict_types=1)` **Adoption**	`100.0%` (`72 / 72`)	`PASS`	Every production file under `Slim/` and test file under `tests/` enforces strict scalar type coercion.
PSR-12 Import Hygiene (`import_hygiene_coverage_pct`)	`98.6%` (`71 / 72`)	`PASS`	Zero wildcard imports, zero late imports; only `1` out-of-order `use RuntimeException;` in `Slim/Handlers/Strategies/RequestResponseNamedArgs.php:16` .
Native Type Safety (`type_safety_coverage_pct`)	`80.1%` (`249 / 311`)	`WARN`	`62` callables omit native parameter type hints on polymorphic `$middleware` / `$toResolve` / `$callable` parameters because PHP union types (`MiddlewareInterface|callable|string`) require PHP `8.0+` while `composer.json` supports PHP `~7.4.0` .
Static Analyzer Annotations	`6` suppressions	`WARN`	`3` `@phpstan-ignore-next-line` (`HttpSpecializedException.php:29` , `ErrorHandler.php:195` , `MiddlewareDispatcher.php:99`) and `3` `@psalm-suppress` (`CallableResolver.php:130` , `MiddlewareDispatcher.php:196` , `RoutingResults.php:94`).
AST PHPDoc Coverage (`docstring_coverage_pct`)	`71.6%`	`WARN`	`0` signature mismatches (`doc_signature_mismatches=0`) and `0` empty docstrings; `98` self-descriptive methods/interfaces omit redundant docblocks.
SIG Complex Functions (`complex_functions_count`)	`3` (`of 311`)	`WARN`	`MiddlewareDispatcher::addDeferred` (`cognitive=26` , `101` LOC), inline anonymous `RequestHandlerInterface::handle` (`cognitive=26` , `68` LOC), and `RouteParser::relativeUrlFor` (`cognitive=13` , `60` LOC).
SIG Parameter Interfacing (`excessive_parameters_count`)	`1` (`of 311`)	`WARN`	Only `Slim\Routing\Route::__construct` (`Slim/Routing/Route.php:120` , `9` parameters) exceeds the SIG 6-parameter threshold.
Global State & Bug-Risk Idioms	`1` static / `1` catch	`PASS`	`CallableResolver::$callablePattern` (`Slim/CallableResolver.php:34` , mutable `public static`) and `MiddlewareDispatcher.php:186` (empty `catch (RuntimeException)` fallback).
12-Factor Operational Readiness (PRR)	`0` env / `0` I/O	`PASS`	`missing_io_timeouts=0` , `scattered_env_access=0` ; `Slim\Logger::log()` (`Slim/Logger.php:30`) uses `error_log(..., 4)` intentionally as the default PSR-3 fallback logger.

Deep Dive: `MiddlewareDispatcher` Complexity & Deferred Resolution

`Slim/MiddlewareDispatcher.php` (`177` LOC) implements Slim's LIFO concentric PSR-15 onion kernel. While `addMiddleware(MiddlewareInterface)` and `addCallable(callable)` are concise closures, `addDeferred(string $middleware)` (`lines 144–244` , `cognitive=26`)

constructs an inline anonymous `RequestHandlerInterface` (`lines 173–240`, `cognitive=26`) to defer PSR-11 container lookup and `'class:method'` parsing until request handling time:

- When `$this->callableResolver instanceof AdvancedCallableResolverInterface`, it calls `$this->callableResolver->resolveMiddleware($this->middleware)` directly.
- Otherwise, it falls back to `$this->callableResolver->resolve($this->middleware)` inside a `try { ... } catch (RuntimeException $e) {}` block (`line 186`), catching and ignoring `RuntimeException` before checking `$this->container->has($callable)` or `class_exists($callable)`. Extracting this inline anonymous class into a dedicated `DeferredMiddlewareHandler` class is the top `[P1]` recommendation in Chapter 3.

5.2 Hermetic Container Runtime Verification (`php:8.2-cli-bookworm` + PCOV `1.0.12`)

PNI501 executed all 4 container runtime verification gates (`RV-01` through `RV-04`) inside an isolated Docker container built from `.pni/dockerfiles/Dockerfile` (`php:8.2-cli-bookworm`, Composer 2, and PECL PCOV `1.0.12`) with network access disabled (`--network=none`).

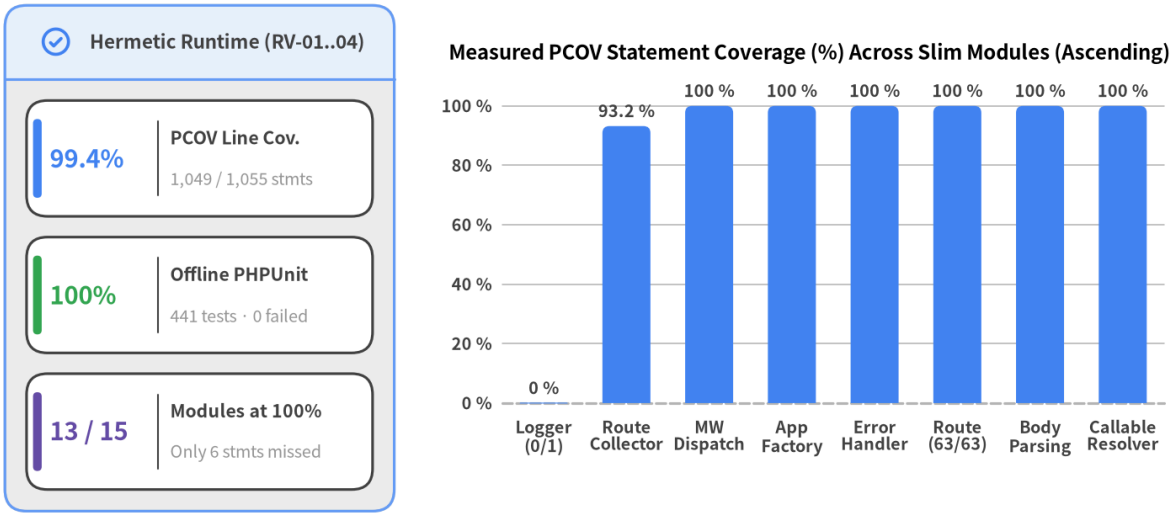


Figure 5.2 — Hermetic PHPUnit + PCOV Line Coverage Across Core Slim Modules (Overall 99.4%: 1,049 / 1,055 Statements)

Runtime Verification Gates (`RV-01` – `RV-04`)

Verification Gate	Status	Duration	Command & Measured Verification Evidence
`RV-01` **Harness Image Build**	`PASSED`	`0.66s`	`.pni/dockerfiles/Dockerfile` (`php:8.2-cli-bookworm` + Composer 2 + PCOV `1.0.12` , `exit_code=0`).
`RV-02` **Air-Gapped PHPUnit Suite**	`PASSED`	`8.56s`	`php -l` syntax check across all `72` `Slim/*.php` files + `phpunit -d pcov.enabled=1` under `--network=none` : `441` **test cases (** `846` **assertions / ** `1,323` **total harness checks) passed**, `0` **failed**, `0` **errors**, `0` **skipped**.
`RV-03` **PCOV Clover Coverage**	`PASSED`	*(with* `RV-02` *)*	`99.4%` **line coverage** (`1,049 / 1,055` core statements covered; `1,052 / 1,058` total Clover lines), with **13 of 15 core modules at** `100.0%` .
`RV-04` **Composer Autoloader Smoke**	`PASSED`	`0.90s`	`php -r "require '/workspace/vendor/autoload.php'; ..."` (`exit_code=0` , `"Composer PSR-4 autoloader verified"`).

Lowest-Covered & Core Modules Breakdown (Ascending Order)

Across the entire Slim codebase, only **6 statements** are unexecuted during the PHPUnit run (`1` in `Slim/Logger.php` and `5` in `Slim/Routing/RouteCollector.php`):

Module Path	Line Coverage	Covered / Total Stmts	Missing Lines	Root-Cause Engineering Explanation
`Slim/Logger.php`	`0.0%`	`0 / 1`	`[30]`	Single `error_log(sprintf('%s: %s', $level, $message), 4)` statement in fallback PSR-3 logger; `ErrorHandlerTest` injects a custom `TestLogger` mock rather than invoking the default `Slim\Logger::log()` .
`Slim/Routing/RouteCollector.php`	`93.2%`	`69 / 74`	`[198, 199, 200, 203, 275]`	Defensive `InvalidArgumentException` / `RuntimeException` guard branches during FastRoute cache file inclusion and route lookup edge cases.
`Slim/MiddlewareDispatcher.php`	`100.0%`	`81 / 81`	`[]`	100% statement coverage across LIFO middleware wrapping, callable adapters, and deferred PSR-11 resolution.
`Slim/Factory/AppFactory.php`	`100.0%`	`77 / 77`	`[]`	100% statement coverage across static app creation and PSR-17 factory auto-discovery.
`Slim/Handlers/ErrorHandler.php`	`100.0%`	`69 / 69`	`[]`	100% statement coverage across HTTP status determination, CORS/Allow header preservation, and `Accept` content negotiation.

Module Path	Line Coverage	Covered / Total Stmts	Missing Lines	Root-Cause Engineering Explanation
`Slim/Routing/Route.php`	`100.0%`	`63 / 63`	`[]`	100% statement coverage across route argument preparation, group middleware chaining, and invocation strategy dispatch.
`Slim/Middleware/BodyParsingMiddleware.php`	`100.0%`	`60 / 60`	`[]`	100% statement coverage across JSON, form-urlencoded, and XML media-type body parsers.
`Slim/CallableResolver.php`	`100.0%`	`58 / 58`	`[]`	100% statement coverage across `'class:method'` regex extraction, PSR-11 container resolution, and invokable objects.
`Slim/App.php`	`100.0%`	`49 / 49`	`[]`	100% statement coverage across `run()`, `handle()`, and middleware registration helpers.
`Slim/ResponseEmitter.php`	`100.0%`	`45 / 45`	`[]`	100% statement coverage via `AdrianSuter\Autoload\Override` interception of `header()`, `headers_sent()`, and `connection_status()`.
`Slim/Error/Renderers/HtmlErrorRenderer.php`	`100.0%`	`44 / 44`	`[]`	100% statement coverage across HTML exception title, description, and stack-trace rendering.
`Slim/Middleware/ErrorMiddleware.php`	`100.0%`	`39 / 39`	`[]`	100% statement coverage across exact-type, subclass-type, and default `Throwable` interception.
`Slim/Routing/RouteParser.php`	`100.0%`	`35 / 35`	`[]`	100% statement coverage across `urlFor()`, `relativeUrlFor()`, and `fullUrlFor()`.
`Slim/Routing/FastRouteDispatcher.php`	`100.0%`	`33 / 33`	`[]`	100% statement coverage across static and variable regex route matching and RFC `Allow` method aggregation.
`Slim/Routing/RouteCollectorProxy.php`	`100.0%`	`29 / 29`	`[]`	100% statement coverage across `get`, `post`, `put`, `patch`, `delete`, `options`, `any`, `group`, and `redirect`.

6. Deep Dive: Security, Supply Chain & API Surface

Slim Framework `v4.15.3` achieves a perfect **100/100 (Grade S)** in `security_vulnerability_audit` (`security` axis: **100/100**; `dependencies` axis: **96/100**). Trivy `0.69.3` supply-chain, secret, and IaC scanning combined with deterministic PHP AST static analysis confirms **zero CVEs across all 93 locked Composer packages** (`7` core runtime + `86` development packages), **zero hardcoded or committed secrets**, **zero container misconfigurations**, and **zero CWE-classified dangerous code sinks** across all `72` PHP source files (`3,149` LOC).

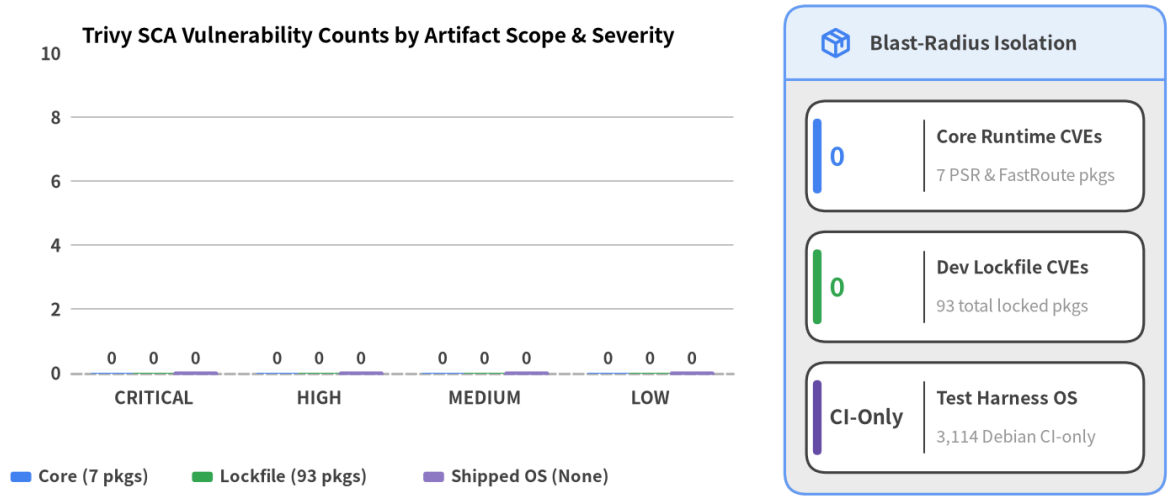


Figure 6.1 — Trivy Vulnerability Isolation Across Slim Core Runtime (7 Pkgs), Composer Lockfile (93 Pkgs), and Ephemeral Test Harness OS

6.1 Composer Supply-Chain Hygiene & Governance Baseline

Slim's `composer.json` and `composer.lock` architecture minimizes downstream supply-chain exposure by keeping the `composer require slim/slim` dependency tree limited strictly to **6 PHP-FIG PSR interface packages** plus `nikic/fast-route ^1.3`, while isolating all 6 PSR-7 concrete test providers (`slim/psr7`, `nyholm/psr7`, `guzzlehttp/psr7`, `laminas/laminas-diactoros`, `httpsoft/http-message`, `slim/http`) and static analyzers under `require-dev`.

Supply-Chain & Security Gate	Measured Value	Verdict	Architectural & Compliance Evidence
Deterministic Lockfile (`has_lockfile`)	`True` (`composer.lock`)	`PASS`	Locks exact commit SHA and version metadata across all `93` runtime and dev packages.
Core Runtime Dependencies (`dependencies_count`)	`7` **packages**	`PASS`	`6` official `psr/*` contract packages + `nikic/fast-route: ^1.3`.
Unpinned Specifiers (`unpinned_dependencies_count`)	`0`	`PASS`	Every dependency uses explicit SemVer caret (`^`) or dual-major union (`^1.0 || ^2.0`) constraints.
Security Disclosure Policy (`has_security_policy`)	`True` (`SECURITY.md`)	`PASS`	Documents responsible vulnerability reporting via GitHub Security Advisories / Tidelift.
Committed / Hardcoded Secrets	`0` **files /** `0` **literals**	`PASS`	Clean across both Trivy secret scanning (`secrets=[]`) and PHP AST literal inspection.
Container IaC Misconfigurations	`0`	`PASS`	`misconfigurations=[]` on `.pni/dockerfiles/Dockerfile` (with batch waiver `DS-0026`).
AST Dangerous Sinks (`ast_dangerous_sinks_count`)	`0`	`PASS`	Zero `eval`, `exec`, `shell_exec`, `system`, `passthru`, `unserialize`, or SQLi/XSS sinks.
Standalone HTTP Endpoints (`api_endpoints_count`)	`0`	`PASS`	Reusable `library` archetype; binds zero network sockets unless instantiated by consumer app.

Separation of Harness OS Packages vs. Library Dependencies

Trivy (`ghcr.io/aquasecurity/trivy:0.69.3`) scanned both the repository filesystem (`composer.lock`) and the resolved container image built from `.pni/dockerfiles/Dockerfile` (`php:8.2-cli-bookworm`):

Vulnerability Boundary	CRITICAL	HIGH	MEDIUM	LOW	Total	Scoring & Consumer Impact
1. Core Runtime Dependencies (`7` Composer pkgs)	`0`	`0`	`0`	`0`	`0`	**Zero Impact**: `psr/*` and `nikic/fast-route` have `0` known advisories.
2. Dev / Test Lockfile (`86` Composer pkgs)	`0`	`0`	`0`	`0`	`0`	**Zero Impact**: All `93` locked Composer packages in `composer.lock` are 100% CVE-free.
3. Shipped Project Container OS (`os_vuln_counts`)	`0`	`0`	`0`	`0`	`0`	**Zero Impact**: Slim is a pure PHP library (`has_dockerfile=false`) and ships no container image.
4. Ephemeral Test Harness OS (`harness_os_vuln_counts`)	`6`	`265`	`1,698`	`1,128`	`3,114`	**Non-Shipped Harness Only**: Debian Bookworm system packages (`perl`, `openssl`, `curl`, `linux-libc-dev`) inside `.pni/dockerfiles/Dockerfile` used solely for offline `phpunit` execution; properly isolated from library scoring.

6.2 SAST Security Verification & Built-In PSR-7 / Error / BodyParsing Defenses

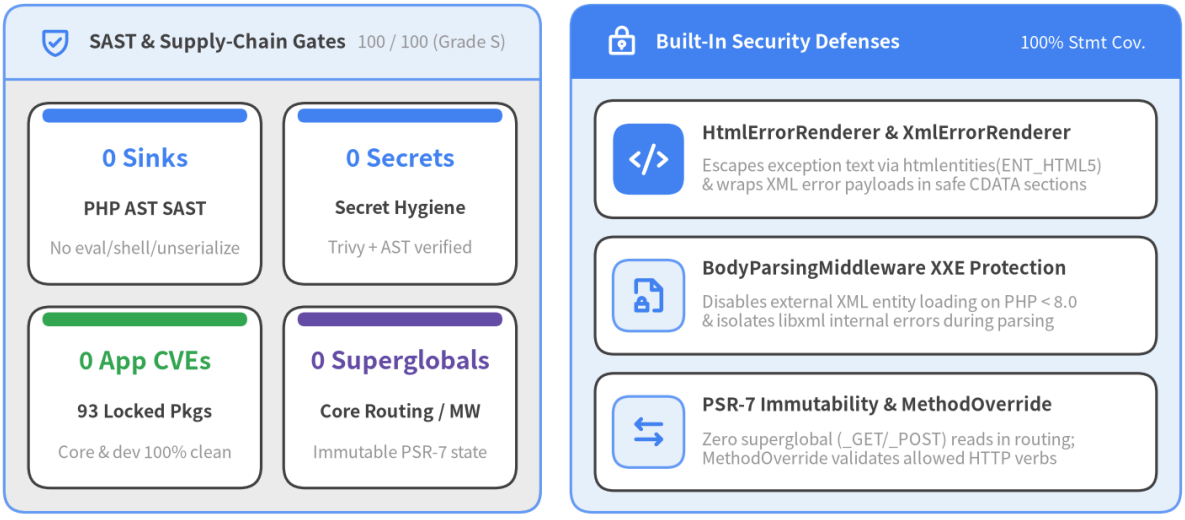


Figure 6.2 — PHP AST SAST Security Scorecard & Built-In Defensive Subsystems (PSR-7, HtmlErrorRenderer, BodyParsingMiddleware)

Architectural Security Analysis of Core Subsystems

1. **Context-Appropriate Output Encoding in `Error/Renderers/*` (`XSS` / `Injection` Prevention)**:
   - `Slim\Error\Renderers\HtmlErrorRenderer` (`Slim/Error/Renderers/HtmlErrorRenderer.php` , `100.0%` PCOV coverage) escapes every exception class name, message, file path, and stack trace via `htmlentities(..., ENT_COMPAT | ENT_HTML5, 'utf-8')` before interpolating into HTML error templates, and gates stack trace disclosure behind the explicit `$displayErrorDetails` boolean flag (default `false`).
   - `Slim\Error\Renderers\XmlErrorRenderer` wraps exception messages inside `<![CDATA[ ... ]]>` blocks while escaping embedded `]]>` sequences, and `JsonErrorRenderer` serializes structured arrays via `json_encode(..., JSON_PRETTY_PRINT | JSON_UNESCAPED_SLASHES)` .
2. **XML External Entity (`XXE`) Hardening in `BodyParsingMiddleware` **:
   - `Slim\Middleware\BodyParsingMiddleware` (`Slim/Middleware/BodyParsingMiddleware.php` , `100.0%` PCOV coverage) guards XML request body parsing (`application/xml` , `text/xml`) by calling `libxml_disable_entity_loader(true)` on PHP `< 8.0` (where external entity loading is not disabled by default) and `libxml_use_internal_errors(true)` with `libxml_clear_errors()` before and after `simplexml_load_string($input)` , preventing XXE file disclosure and warning leakage.
3. **Immutable PSR-7 Request State & Superglobal Isolation**:
   - Core routing and middleware classes never read `$_GET` , `$_POST` , `$_SERVER` , or `$_COOKIE` directly. Superglobals are read exclusively once inside `ServerRequestCreatorFactory` when synthesizing the initial PSR-7 `ServerRequestInterface` .
   - `RoutingMiddleware` attaches matched `RouteContext` state (`__route__` , `__routeParser__` , `__routingResults__` , `__basePath__`) via immutable `$request->withAttribute(...)` clones, and `ResponseEmitter::emitHeaders()` preserves multiple `Set-Cookie` headers without overwriting prior cookie directives.